Dr. Vishwanath Karad
MIT WORLD PEACE UNIVERSITY | PUNE
TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

B. Tech. Electronics and Communication Engineering (First Year) (Batch 2018-19)

Trimester – I

Sr. No.	Course Code	Name of Course	Type	Weekly Workload, Hrs.			Credits		Assessment, Marks			
				Theory	Tutorial	Lab.	Th.	Lab.	CCA*	LCA*	ETT	Total
1		Mathematics -I	BS	3	1	--	3	--	100	-	50	150
2		Physics	BS	3	1	2	3	1	100	50	50	200
3		Applied Mechanics	ES	3	1	2	3	1	100	50	50	200
4		Workshop Practices	ES	-	-	2	-	1	-	50	-	50
5		Effective Communication	HSS	2	-	2	1	1	50	50	-	100
6		Practicing Yoga and Meditation	WP	-		1	-	-	-	-	-	-
		Total :		**11**	**3**	**9**	**10**	**4**	**350**	**200**	**150**	**700**

Type :(Refer para 11 of Academic Ord. 2017)

Weekly Teaching Hours: 23 Hours

Total Credits: First Year B. Tech ECE Trimester - I: 14

**Assessment Marks are valid only if Attendance criteria are met

* CCA: Class Continuous Assessment

* LCA: Laboratory Continuous Assessment

Prof. (Dr.) L. K. Kshirsagar
Dean

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B. Tech. Electronics and Communication Engineering (First Year) (Batch 2018-19)

Trimester – II

Sr. No.	Course Code	Name of Course	Type	Weekly Workload, Hrs.			Credits		Assessment, Marks			
				Theory	Tutorial	Lab.	Th.	Lab.	CCA*	LCA*	ETT	Total
1		Mathematics II	BS	3	1	-	3	-	100	-	50	150
2		Biology	BS	3	-	-	2	-	50	-	50	100
3		Material Science For Engineers	BS	3	-	2	2	1	50	50	50	150
4		Engineering Graphics	ES	2	-	2	1	1	-	50	50	100
5		Computer Programming	ES	3	-	2	2	1	50	50	50	150
6		Indian Culture and Heritage	WP	3	-	-	2	-	50	-	50	100
7		Rural Immersion Programme	WP	-	-	-	-	-	-	-	-	-
8		Practicing Yoga and Meditation	WP	-	-	1	-	-	-	-	-	-
		Total :		**17**	**1**	**7**	**12**	**3**	**300**	**150**	**300**	**750**

Type :(Refer para 11 of Academic Ord. 2017)

Weekly Teaching Hours: 25 Hours

Total Credits: First Year B. Tech ECE Trimester - II: 15

**Assessment Marks are valid only if Attendance criteria are met

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* LCA: Laboratory Continuous Assessment

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B. Tech. Electronics and Communication Engineering (First Year) (Batch 2018-19)

Trimester – III

Sr. No.	Course Code	Name of Course	Type	Weekly Workload, Hrs.			Credits		Assessment Marks**			
				Theory	Tutorial	Lab.	Th.	Lab.	CCA*	LCA*	ETT	Total
1		Chemistry	BS	3	-	2	2	1	50	50	50	150
2		Engineering Design Principles	ES	3	-	2	2	1	50	50	50	150
3		Engineering Science Elective Course I*	ES	3	-	2	2	1	50	50	50	150
4		Engineering Science Elective Course II*	ES	3	-	2	2	1	50	50	50	150
5		Philosophers, Great Kings & Dynasties	WP	3	-	-	2	-	50	-	50	100
6		Practicing Yoga and Meditation	WP	-	-	1	-	-	-	-	-	-
		Total :		**15**	**-**	**9**	**10**	**4**	**250**	**200**	**250**	**700**

Type :(Refer para 11 of Academic Ord. 2017)

Weekly Teaching Hours: 24 Hours

Total Credits: First Year B. Tech EE Trimester - III: 14

Total First Year B. Tech Credits: 14+15+14=43

**Assessment Marks are valid only if Attendance criteria are met

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* LCA: Laboratory Continuous Assessment

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Dean

FY B. Tech. Electives:

Any two courses other than parent/ home discipline can be chosen from the list given below:

Sr. No.	Course Code	Name of Course
1		Introduction to Civil Engineering
2		Introduction to Mechanical Engineering
3		Introduction to Computer Science and Engineering
4	EE121	Introduction to Electrical Engineering
5	EC131	Introduction to Electronics Engineering
6		Introduction to Polymer Engineering
7		Introduction to Petroleum Engineering
8		Introduction to Chemical Engineering

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B. Tech. Electronics and Communication Engineering (Second Year) (Batch 2018-19)

Trimester – IV

Sr. No.	Course Code	Name of Course	Type	Weekly Workload, Hrs.			Credits		Assessment Marks**			
				Theory	Tutorial	Lab.	Th.	Lab.	CCA*	LCA*	ETT	Total
1	SCI201A	Mathematics-III	BS	3	1	--	2	1	100	--	50	150
2	ECE201A	Electronic Devices and Circuits	ES	4	--	2	3	1	100	50	50	200
3	ECE202A	Digital Electronics	PC	3	--	2	2	1	50	50	50	150
4	ECE203A	Computer Organization and Microprocessors	PC	3	--	2	2	1	50	50	50	150
5	IC	Indian Constitution	HSS	2	--	--	1	--	50	--	--	50
		Total:		**15**	**1**	**6**	**10**	**4**	**350**	**150**	**200**	**700**

****Assessment Marks are valid only if Attendance criteria are met**

Weekly Teaching Hours: 22 Hours

Total Credits: Second year B. Tech ECE Trimester - I: 14

* CCA: Class Continuous Assessment

* LCA: Laboratory Continuous Assessment

B. Tech. Electronics and Communication Engineering (Second Year) (Batch 2018-19)

Trimester – V

Sr. No.	Course Code	Name of Course	Type	Weekly Workload, Hrs.			Credits		Assessment Marks**			
				Theory	Tutorial	Lab.	Th.	Lab.	CCA*	LCA*	ETT	Total
1	ECE205A	Signals and Systems	PC	4	--	--	3	--	100	--	50	150
2	ECE206A	Linear Integrated Circuits	PC	3	--	2	2	1	50	50	50	150
3	ECE207A	Data Structures and algorithms	PC	3	--	2	2	1	50	50	50	150
4	ECE208A	Hardware/Software tools for Electronics Engineering	PC	--	--	2	--	1	--	50	--	50
5	EEE213A	Electrical Circuits and Machines	PC	3	--	2	2	1	50	50	50	150
6	WPC501A	Philosophy of science and religion/spirituality	WP	3	--	--	2	--	70	--	30	100
7		National Study tour	WP	--	--	--	--	--	--	--	--	--
		Total:		**16**	--	**8**	**11**	**4**	**320**	**200**	**230**	**750**

****Assessment Marks are valid only if Attendance criteria are met**

Weekly Teaching Hours: 24 Hours

Total Credits: Second year B. Tech ECE Trimester - II: 15

* CCA: Class Continuous Assessment

* LCA: Laboratory Continuous Assessment

B. Tech. Electronics and Communication Engineering (Second Year) (Batch 2018-19)

Trimester – VI

Sr. No.	Course Code	Name of Course	Type	Weekly Workload, Hrs.			Credits		Assessment Marks**			
				Theory	Tutorial	Lab.	Th.	Lab.	CCA*	LCA*	ETT	Total
1	ECE212A	Analog Communication	PC	3	--	2	2	1	50	50	50	150
2	ECE213A	Object Oriented Programming	PC	3	--	2	2	1	50	50	50	150
3	ECE214A	Microcontrollers	PC	4	--	2	3	1	50	50	50	200
4	EEE207A	Control Systems	PC	3	--	2	2	1	100	50	50	150
5	HSS01	Environmental Science	HSS	2	--	--	1	--	50	--	--	50
		Total:		**15**	--	**8**	**10**	**4**	**300**	**200**	**200**	**700**

****Assessment Marks are valid only if Attendance criteria are met**

Weekly Teaching Hours: 23 Hours

Total Credits: Second year B. Tech ECE Trimester - III: 14

* CCA: Class Continuous Assessment

* LCA: Laboratory Continuous Assessment

Total Second Year B. Tech Credits: 14+15+14= 43

B. Tech. Electronics and Communication Engineering (Third Year) (Batch 2018-19)

Trimester – VII

Sr. No.	Course Code	Name of Course	Type	Weekly Workload, Hrs.			Credits		Assessment Marks**			
				Theory	Tutorial	Lab.	Th.	Lab.	CCA*	LCA*	ETT	Total
1	EC311	Digital Signal Processing	PC	4	-	2	2	1	50	50	50	150
2	EC312	Digital Communication	PC	4	-	2	2	1	50	50	50	150
3	EC313	System Programming and RTOS	PC	4	-	2	2	1	50	50	50	150
4	EC314	Computer Networks	PC	4	-	2	2	1	50	50	50	150
5	EC315	Hardware Maintenance & Programming skills for Electronic software	PC	-	-	2		1		50		50
6	WPC701A	Indian Tradition Culture and Heritage		3			2		70		30	100
		Total:		**19**	**-**	**10**	**10**	**5**	**270**	**250**	**230**	**750**

****Assessment Marks are valid only if Attendance criteria are met**

Weekly Teaching Hours: 29 Hours

Total Credits: Third Year B. Tech ECE Trimester - I: 15

* CCA: Class Continuous Assessment

* LCA: Laboratory Continuous Assessment

B. Tech. Electronics and Communication Engineering (Third Year) (Batch 2018-19)

Trimester – VIII ***

Sr. No.	Course Code	Name of Course	Type	Weekly Workload, Hrs.			Credits		Assessment Marks**			
				Theory	Tutorial	Lab.	Th.	Lab.	CCA*	LCA*	ETT	Total
1	ECPE1	Professional Elective -I	PE	4	-	2	2	1	50	50	50	150
2	ECPE2	Professional Elective -II	PE	4	-	2	2	1	50	50	50	150
3	ECOE1	Open Elective-I	OE	4	-	-	2	-	50		50	100
4	ECOE2	Open Elective-II	OE	4	-	-	2	-	50		50	100
5	ECTE1	Mini Project-I	PBL		-	4	-	2	-	100	-	100
		Total:		**16**	**-**	**8**	**8**	**4**	**200**	**200**	**200**	**600**

****Assessment Marks are valid only if Attendance criteria are met**

Weekly Teaching Hours: 24 Hours

* CCA: Class Continuous Assessment

Total Credits: Third Year B. Tech ECE Trimester - II: 12

* LCA: Laboratory Continuous Assessment

*** **Credits for internship will be 12, in lieu of the above subjects (1-5), if the student opts for internship. Internship will be opted for only one trimester out of (VIII, IX & X)**

The subjects related to **Peace (Humanities: Ethical, Moral and Social Sciences) and Finance (Finance and Costing)** may be repeated for three trimesters (3/2, 3/3, 4/1), for 2 credits each, so that it will be easy for students to attempt them in any of the trimester with flexibility.

B. Tech. Electronics and Communication Engineering (Third Year) (Batch 2018-19)

Trimester – IX ***

Sr. No.	Course Code	Name of Course	Type	Weekly Workload, Hrs.			Credits		Assessment Marks**			
				Theory	Tutorial	Lab.	Th.	Lab.	CCA*	LCA*	ETT	Total
1	ECPE3	Professional Elective -III	PE	4	-	2	2	1	50	50	50	150
2	ECPE4	Professional Elective -IV	PE	4	-	2	2	1	50	50	50	150
3	ECOE3	Open Elective-III	OE	4	-	-	2	-	50		50	100
4	ECOE4	Open Elective-IV	OE	4	-	-	2	-	50		50	100
5	ECTE2	Mini Project-II	PBL		-	4	-	2	-	100	-	100
		Total:		**16**	**-**	**8**	**8**	**4**	**200**	**200**	**200**	**600**

****Assessment Marks are valid only if Attendance criteria are met**

Weekly Teaching Hours: 24 Hours

Total Credits: Third Year B. Tech ECE Trimester - III: 12

* CCA: Class Continuous Assessment

* LCA: Laboratory Continuous Assessment

*** **Credits for internship will be 12, in lieu of the above subjects (1-5), if the student opts for internship. Internship will be opted for only one trimester out of (VIII, IX & X)**

Students completing internship in Tri VIII will have to undergo a bridge course for prerequisite courses required for professional electives only. One prerequisite course will have a CCA of 100 marks mapped to 2 credits.

The subjects related to **Peace (Humanities: Ethical, Moral and Social Sciences) and Finance (Finance and Costing)** may be repeated for three trimesters (3/2, 3/3, 4/1), for 2 credits each, so that it will be easy for students to attempt them in any of the trimester with flexibility.

Total Credit for Third year B.Tech = 15 +12 +12 = 39

B. Tech. Electronics and Communication Engineering (Fourth Year) (Batch 2018-19)

Trimester –X ***

Sr. No.	Course Code	Name of Course	Type	Weekly Workload, Hrs.			Credits		Assessment Marks**			
				Theory	Tutorial	Lab.	Th.	Lab.	CCA*	LCA*	ETT	Total
1	PEC5	Professional Elective –V	PE	4	-	2	2	1	50	50	50	150
2	PEC6	Professional Elective –VI	PE	4	-	2	2	1	50	50	50	150
3	OEC5	Open Elective-V	OE	4	-	-	2	-	50		50	100
4	OEC6	Open Elective-VI	OE	4	-	-	2	-	50		50	100
5	TE3	Mini Project-II	PBL		-	4	-	2	-	100	-	100
		Total:		**16**	**-**	**8**	**8**	**4**	**200**	**200**	**200**	**600**

****Assessment Marks are valid only if Attendance criteria are met**

Weekly Teaching Hours: 24 Hours

Total Credits: Fourth Year B. Tech ECE Trimester - I: 12

* CCA: Class Continuous Assessment

* LCA: Laboratory Continuous Assessment

*** **Credits for internship will be 12, in lieu of the above subjects (1-5), if the student opts for internship. Internship will be opted for only one trimester out of (VIII, IX & X).** Students completing internship in Tri IX will have to undergo a bridge course for prerequisite courses required for professional electives only. One prerequisite course will have a CCA of 100 marks mapped to 2 credits. The subjects related to **Peace (Humanities: Ethical, Moral and Social Sciences) and Finance (Finance and Costing)** may be repeated for three trimesters (3/2, 3/3, 4/1), for 2 credits each, so that it will be easy for students to attempt them in any of the trimester with flexibility.

B. Tech. Electronics and Communication Engineering (Fourth Year) (Batch 2018-19)

Trimester –XI

Sr. No.	Course Code	Name of Course	Type	Weekly Workload, Hrs.			Credits		Assessment Marks**			
				Theory	Tutorial	Lab.	Th.	Lab.	CCA*	LCA*	ETT	Total
1	ECPP1A1	Project Management	HSS	4			3		150			150
2	ECPP1A2	Capstone Project Phase-1	PBL			10		5		250		250
		Total:		**4**		**10**	**3**	**5**	**150**	**250**		**400**

Weekly Teaching Hours: 14 Hours

Total Credits: Fourth Year B. Tech ECE Trimester - II: 8

****Assessment Marks are valid only if Attendance criteria are met**

* CCA: Class Continuous Assessment

* LCA: Laboratory Continuous Assessment

During Capstone project a course on Project management for 3 credits needs to be completed as a part of the same.
One web based course can be started in Tri XI. Credits for web based course to be added in last trimester.

B. Tech. Electronics and Communication Engineering (Fourth Year) (Batch 2018-19)

Trimester –XII***

Sr. No.	Course Code	Name of Course	Type	Weekly Workload, Hrs.			Credits		Assessment Marks**			
				Theory	Tutorial	Lab.	Th.	Lab.	CCA*	LCA*	ETT	Total
1	MOOC1	Web Based Course 1	Self Study				2		100			100
2	MOOC2	Web Based Course 2	Self Study				2		100			100
3	WPC111A	Scientific Studies of Mind, Matter, Spirit and Consciousness	WPC	3			2		70		30	100
4	ECCPP2	Capstone Project Phase-2	PBL			20		10		500		500
		Total:		**3**		**20**	**6**	**10**	**270**	**500**	**30**	**800**

****Assessment Marks are valid only if Attendance criteria are met**

Weekly Teaching Hours: 23 Hours

Total Credits: Fourth Year B. Tech ECE Trimester - III: 16

* CCA: Class Continuous Assessment

* LCA: Laboratory Continuous Assessment

Total Credit for Fourth year B.Tech = 12 +8 +16 = 36

Total credit for B.Tech Program = 43 + 43 + 39 + 36 + 2(Peace) +2 (Finance) =165

LIST OF PROFESSIONAL ELECTIVES OFFERED BY SCHOOL OF ELECTRONICS AND COMMUNICATION ENGINEERING

BASKET OF PROFESSIONAL ELECTIVES					
BASKET NO.	**BASKET -1**	**BASKET -2**	**BASKET -3**	**BASKET-4**	**BASKET-5**
Name of Basket	**Communication Engineering**	**Signal and Image Processing**	**VLSI & Embedded**	**Artificial Intelligence**	**Electronics Engineering**
ECPE1	Electromagnetics	Advanced DSP	FPGA & ASIC design	Maths for Artificial Intelligence	Power Electronics
ECPE2	Coding Techniques	Digital Image Processing	Embedded Processors	Fuzzy Logic & Graph Theory	Mechatronics
ECPE3	Antenna Design	Computer Vision	CMOS Digital VLSI design	Optimization Techniques	Electronics Product Design
ECPE4	Wireless Communication	Pattern Recognition	Electronics Product Design	Machine Learning	Industrial Automation
ECPE5	Microwave and Satellite Communication	Digital Video Processing	Mixed Signal CMOS design	Artificial Neural Networks	Atmospheric Instrumentation
	Wireless Sensor Network	Speech Processing			Agriculture Electronics
	Network Protocols				
ECPE6	Fiber Optics Communication	Bio signal Processing	Automotive Electronics	Bioinformatics	Biomedical Electronics
	Internet of Things	Natural Language Processing		Natural Language Processing	Robotics and Automation
	Software Defined Radios	AI for Medical Imaging		AI for Medical Imaging	
	Network Security				